

Sharp Euler Bounds and Uniform Asymptotics in Polylogarithmic Analysis

Reflected moments, Lerch zeros, and integral shuffle lattices

A research continuation for the ProveIt project

10 October 2026

Abstract

We continue the polylogarithm manuscript at the pinned repository revision identified in Appendix A. Four proof developments are obtained. First, an elementary multiplicative-increment inequality controls every Gaussian Euler remainder at arbitrary positive real orders. It identifies the least universal constant as $C_* = \max_{b>0} \{\beta(b) + 2^{-b}\eta(b)\}$, and exact rational certificates enclose its unique maximizer and its value. Second, the coordinate $y = \log \Gamma(x)$ gives an all-orders expansion of reflected log-gamma moments uniformly for every ratio of their two nonnegative exponents, as the larger exponent tends to infinity. Third, an explicit Appell-root separation bound and a logarithmic variation estimate for the Lerch kernel yield an effective saturation threshold and simultaneous zero locations when the Laurent index grows with the derivative order. Fourth, a central-factorial identity integrally diagonalizes the odd-weight shuffle minor, giving its cokernel, Smith factors, field ranks, and optimal common inverse denominator. Weight-eleven and weight-thirteen Gaussian relations are recorded as conjectures with exact finite proximity certificates. Established source results, new proofs, finite certificates, numerical diagnostics, and unresolved claims are distinguished throughout.

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1 Scope, prior results, and the new claims

The starting point is the 375-page manuscript in `Analysis/Polylogarithms/docs/manuscript`, together with the relevant reports already present in the same repository at the pinned revision [1]. The version matters: the S_4 identity, several Gaussian reductions, and the rational relation ranks were already proved there. Repeating those results as new conjecture resolutions would misstate the state of the project. The existing S_6 special-value conjecture remains unresolved by the present work.

The objective is to close specific gaps in the current programme and provide reusable structural results. The universal Euler question has a particularly direct answer: the existing Gaussian-axis maximum really is the sharp error constant, once a new comparison is proved for *all* Euler indices. The reflected-moment question requires a different kind of step. Separate fixed-exponent and bounded-ratio asymptotics do not justify substitution of an unbounded exponent ratio. A change of coordinate and an appropriate placement of the Jacobian in the phase give uniform derivative and tail estimates on the full parameter range.

The zero problem has a similar obstruction to naive extrapolation. A fixed-index expansion on a compact scale does not control the extreme roots when the index grows. Our proof works with relative polynomial errors on the logarithmic coordinate and with a scale-independent positive-weight inequality. The resulting threshold is intentionally conservative, but it is explicit, uniform in the Lerch parameter, and applies to every positive root at once.

The algebraic part is separate from those analytic estimates. A rational rank says that a coordinate system works over $\mathbb{Q}$. Integral diagonalization additionally determines its torsion and all primes of coordinate failure. The matrix is closely related to Zagier’s double-zeta matrix [7, 8, 9]; this connection is credited rather than treated as evidence of new priority. The proof given here is constructive and supplies the integral information directly in the manuscript’s shuffle coordinates.

1.1 A claim and dependency map

Result	New conclusion	Inherited input or limitation
Euler remainder	The least constant uniform in $a, b > 0$ and $N \geq 1$ is C_* ; $C(b)$ is optimal at each fixed b .	The source’s unique-axis-maximum theorem identifies the sole maximizing b_* .
Exact maximum	Rational enclosures for b_* and C_* , and the useful budget $C_* < 57/50$.	Finite exact arithmetic plus the proved qualitative unimodality.
Reflected moments	All fixed asymptotic orders, uniformly for $0 \leq m \leq n$ as $n \rightarrow \infty$.	The previously certified global log-gamma slope inequality; no exponentially small error is claimed.
Lerch zeros	Explicit K_n and relative locations uniform in every branch and $0 \leq \rho \leq 1$; a growing-index regime follows.	The standard positive integral and Laplace zero-count principle, reviewed in the proof. The threshold is sufficient, not sharp.

Result	New conclusion	Inherited input or limitation
Shuffle lattice	Integral equivalence to $\text{diag}(1, 3, \dots, 2m - 1)$, hence the full arithmetic of the minor.	A formal shuffle presentation; no period-independence conclusion.
Gaussian candidates	Frozen S_{10} and S_{12} formulas with exact normalized residual bounds below 10^{-650} .	They remain conjectures; the residual intervals contain zero.

All main analytic and algebraic statements are proved in ordinary mathematical language. The exact certificates have explicitly narrower roles: they certify constant brackets, check a pre-existing slope inequality, or establish finite residual enclosures. They are not a substitute for an identity proof, and no Lean formalization is claimed.

1.2 Notation and branch conventions

We use decreasing nested indices:

$$\text{Li}_{a,b}(z, 1) = \sum_{n>r\geq 1} \frac{z^n}{n^a r^b}, \quad H_n^{(b)} = \sum_{r=1}^n r^{-b}, \quad H_n = H_n^{(1)}.$$

The double family $F_{a,b}(z)$ always denotes this one-variable specialization, initially in $|z| < 1$. Real-order values at i mean the principal analytic continuation. At some small real orders the ordinary boundary series does not converge, which does not affect the finite Euler transforms or their proved limit. All gamma powers in real integrals use positive real bases. The Dirichlet beta and eta functions are denoted by β and η ; Euler's constant is γ .

In the moment section only, $F(x) = -\psi(x)$ is a positive real function; the argument and absence of subscripts distinguish it from the double polylogarithm. In the Lerch section $F_{n,k}^\rho$ is a signed normalization of a parameter derivative, defined explicitly there. These local notations follow the corresponding source problems and are not identifications of different functions.

1.3 Reading the computational evidence

The accompanying archive contains the full LaTeX source, compiled PDF, reusable section fragments, exact verifiers, frozen candidates, interval certificates, numerical diagnostics, figure-generation code, a provenance record, and integration notes. Each computational result says what it establishes. For example, the equality $UMV = D$ is exact integer algebra; an enclosure $|R| < 10^{-650}$ is a finite analytic error statement; and a plot of asymptotic errors is a floating-point diagnostic. None of these labels is used interchangeably.

The original log-gamma moment problem and its fixed-order asymptotics have substantial prior literature [2, 3]. Our contribution is the unrestricted-ratio theorem, rather than the definition or first asymptotic observation. Likewise, the classical Laplace and gamma-series tools used below are standard [4, 5]; uniformity in the present parameter domain is proved explicitly.

2 The sharp universal Euler constant at all positive real orders

Write

$$c_n = \frac{H_{n-1}^{(b)}}{n^a}, \quad F_{a,b}(z) = \sum_{n \geq 1} c_n z^n, \quad a \geq 0, \quad b > 0,$$

where values away from the disk use the principal continuation to $\mathbb{C} \setminus [1, \infty)$. Set

$$g_{a,b} = \operatorname{Im} F_{a,b}(i), \quad d_j = \sum_{k=0}^j (-1)^k \binom{j}{k} c_{2k+1}, \quad E_N = \sum_{j=0}^{N-1} 2^{-j-1} d_j \quad (N \geq 1).$$

The manuscript proves one-sided Euler convergence throughout this domain, but explicitly leaves a universal error constant in the supercritical fractional-order region open. The Gaussian maximum established there controls only the first Euler remainder. The following scalar inequality supplies the missing comparison for every later remainder.

2.1 An elementary comparison of scalar Euler kernels

For an integer $N \geq 1$ put

$$\Phi_N(t) = \frac{(1-t)^N}{1+t} \quad (0 \leq t \leq 1).$$

Lemma 2.1 (Uniform multiplicative-increment bound). *For every integer $N \geq 1$ and $q, t \in [0, 1]$,*

$$0 \leq \Phi_N(qt) - \Phi_N(t) \leq \frac{1-q}{1+q}. \quad (1)$$

The upper inequality is strict when $0 < q < 1$ and $0 < t < 1$. For $0 < q < 1$ and $t = 1$, equality in the upper inequality occurs exactly when $N = 1$.

Proof. The first inequality follows from the decrease of Φ_N . For the second define $f_N(u) = u^N/(2-u)$ on $[0, 1]$ and set $A = 1 - qt$, $B = 1 - t$, and $p = A + B$. The cases $q = 1$ or $t = 0$ are immediate, so suppose $A > B$. The desired inequality is equivalent to

$$\frac{f_N(A) - f_N(B)}{A - B} \leq \frac{1}{2 - A - B}. \quad (2)$$

The exceptional case $A = B = 1$ has already been excluded.

The derivative f'_N is strictly convex on $(0, 1)$: expand

$$f_N(u) = \sum_{j=0}^{\infty} \frac{u^{N+j}}{2^{j+1}}$$

and differentiate three times; the result is strictly positive for $0 < u < 1$. The average of a convex function over a centered interval is nondecreasing as its length increases. Explicitly, if $h \mapsto (2h)^{-1} \int_{c-h}^{c+h} g(v) dv$ is differentiated, the result is nonnegative by the trapezoid bound for a convex g . It is strictly positive for a strictly convex g and a nondegenerate interval. Applying this to $g = f'_N$ shows that the divided difference on the left of (2), at fixed sum p , increases as the endpoints move apart.

If $p \leq 1$, enlarge $[B, A]$ to $[0, p]$. Then

$$\frac{f_N(A) - f_N(B)}{A - B} \leq \frac{f_N(p) - f_N(0)}{p} = \frac{p^{N-1}}{2-p} \leq \frac{1}{2-p}.$$

If $p \geq 1$, enlarge it instead to $[p-1, 1]$. Since $f_N(1) = 1$ and $f_N(p-1) \geq 0$,

$$\frac{f_N(A) - f_N(B)}{A - B} \leq \frac{1 - f_N(p-1)}{2-p} \leq \frac{1}{2-p}.$$

When $0 < t < 1$ and $0 < q < 1$, both original endpoints are interior to $[0, 1]$, so the interval is strictly enlarged in the appropriate case and strict convexity gives a strict inequality. At $t = 1$ the original difference equals $(1-q)^N/(1+q)$, proving the equality statement. The endpoint $q = 0$ follows directly from $\Phi_N(0) - \Phi_N(t) = 1 - \Phi_N(t) \leq 1$. $\square$

2.2 A probability formula for every Euler remainder

For $a > 0$ let $S \sim \text{Gamma}(a, 1)$ and $T \sim \text{Gamma}(b, 1)$ be independent, and set $X = e^{-S}$, $V = e^{-T}$. When $a = 0$, define $S = 0$, $X = 1$ deterministically. A finite geometric sum and the Gamma integral give

$$c_n = \mathbb{E} \left[\frac{X^{n-1} - (XV)^{n-1}}{1 - V} \right] \quad (n \geq 1). \quad (3)$$

Indeed the expectation factors as $n^{-a} \sum_{k=0}^{n-2} (k+1)^{-b}$, including the empty sum at $n = 1$. For $a > 0$ this is the existing positive double kernel in probability coordinates; no signed moment measure is required.

Theorem 2.2 (Universal one-sided remainder identity and domination). *For every $a \geq 0$, $b > 0$ and integer $N \geq 1$, the Euler series converges absolutely, $d_0 = 0$, $d_j < 0$ for $j \geq 1$, and E_N decreases strictly to $g_{a,b}$. More precisely, with $R_N(a, b) = 2^N(E_N - g_{a,b})$,*

$$R_N(a, b) = \mathbb{E} \left[\frac{\Phi_N(X^2 V^2) - \Phi_N(X^2)}{1 - V} \right] > 0. \quad (4)$$

Define

$$C(b) = \beta(b) + 2^{-b} \eta(b) = \mathbb{E} \left[\frac{1 + V}{1 + V^2} \right], \quad (5)$$

where β is the Dirichlet beta function and $\eta(b) = \sum_{n \geq 1} (-1)^{n-1} n^{-b}$. Then

$$R_N(a, b) < C(b) \quad (a > 0, b > 0, N \geq 1). \quad (6)$$

At $a = 0$, the same estimate is non-strict, and equality holds exactly when $N = 1$. In addition, $R_N(a, b) \rightarrow 0$ as $N \rightarrow \infty$ for every fixed pair in the stated domain.

Proof. Applying a finite binomial sum to (3) gives

$$d_j = \mathbb{E} \left[\frac{(1 - X^2)^j - (1 - X^2 V^2)^j}{1 - V} \right].$$

This is zero for $j = 0$ and strictly negative for $j \geq 1$. Every fixed integrand is bounded by $2j$, by the mean-value theorem, so all displayed finite expectations are legitimate.

The positive double-kernel continuation at i gives

$$-g_{a,b} = \mathbb{E} \left[\frac{X^2(1+V)}{(1+X^2)(1+X^2V^2)} \right] > 0. \quad (7)$$

This remains valid on the axis $a = 0$ from $F_{0,b}(z) = z \operatorname{Li}_b(z)/(1-z)$. Alternatively, expanding the two denominators in the disk gives (3), and dominated continuation gives (7).

Tonelli's theorem, applied to the nonnegative functions $-d_j$, permits summing the infinite Euler series before proving its finiteness. The geometric sum gives

$$\sum_{j \geq 0} \frac{-d_j}{2^{j+1}} = \mathbb{E} \left[\frac{(1+X^2V^2)^{-1} - (1+X^2)^{-1}}{1-V} \right] = -g_{a,b} < \infty.$$

Thus the Euler series is absolutely convergent and its sum is g . The same geometric summation beginning at $j = N$ proves (4). The positivity and strict decrease of E_N follow from the strict signs of d_j for $j \geq 1$.

Apply Lemma 2.1 with $t = X^2$ and $q = V^2$. It gives the pointwise domination

$$0 < \frac{\Phi_N(X^2V^2) - \Phi_N(X^2)}{1-V} \leq \frac{1-V^2}{(1+V^2)(1-V)} = \frac{1+V}{1+V^2}.$$

The upper inequality is strict almost surely for $a > 0$. On the axis it is equality precisely at $N = 1$. Taking expectations proves (6). To identify $C(b)$, write the density of V as $(-\log v)^{b-1} dv/\Gamma(b)$ on $(0,1)$, and integrate the even and odd geometric series for $(1+v)/(1+v^2)$. Their partial sums are bounded uniformly, justifying dominated convergence; the resulting alternating Dirichlet series are exactly $\beta(b) + 2^{-b}\eta(b)$ for every $b > 0$.

For every $0 < X \leq 1$ and $0 < V < 1$, both powers in the numerator of (4) tend to zero as $N \rightarrow \infty$. The same integrable domination, independent of N , proves $R_N \rightarrow 0$. $\square$

2.3 The least constant uniform in both real orders

The preceding manuscript already proves that C has exactly one stationary point b_* , a nondegenerate global maximum, with $1 < b_* < 2$. Denote its value by

$$C_* := C(b_*) = \max_{b>0} \{\beta(b) + 2^{-b}\eta(b)\}.$$

Its reported numerical diagnostics are

$$b_* \approx 1.30221658710124120959, \quad C_* \approx 1.13656110333950956095;$$

these decimal values are not needed for the theorem. The existing exact bounds $227/200 < C_* < (1 + \sqrt{2})/2$ may be retained.

Corollary 2.3 (Sharp universal real-order Euler bound). *For all $a, b > 0$ and all integers $N \geq 1$,*

$$E_N - C_* 2^{-N} < g_{a,b} < E_N. \quad (8)$$

The number C_ is the least constant for which this assertion holds uniformly over the entire positive quadrant. With $a = 0$ included, $0 < R_N \leq C_*$, and equality occurs exactly at $(a, b, N) = (0, b_*, 1)$. For each fixed $b > 0$, the least constant uniform in $a > 0$ and $N \geq 1$ is $C(b)$.*

Proof. The upper bound follows from (6) and $C(b) \leq C_*$. For sharpness, take $N = 1$, for which $E_1 = 0$ and $R_1 = -2g$. As $a \downarrow 0$, $S \sim \text{Gamma}(a, 1)$ tends to zero in probability (its mean is a), so $X \rightarrow 1$. The bounded integrand in (7), or the pointwise domination already proved, gives

$$\lim_{a \downarrow 0} R_1(a, b) = C(b).$$

Taking $b = b_*$ shows that every smaller universal constant fails for some positive a . The same argument at a fixed b proves its claimed sharp constant. Strictness and the sole equality case on the closed outer-order axis follow from the theorem and the uniqueness of the maximum of C . $\square$

The derivative of $(1+v)/(1+v^2)$ is $(1-2v-v^2)/(1+v^2)^2$, so its maximum on $[0, 1]$ is $(1+\sqrt{2})/2 < 5/4$, attained at $v = \sqrt{2} - 1$. Theorem 2.2 therefore supplies the self-contained budget $R_N(a, b) < 5/4$ throughout the domain. The explicit algebraic choice $(1+\sqrt{2})/2$ gives an immediate universal enclosure without optimization in b . The established constant one for $a \geq 1$ remains sharper on that restricted region. The new result resolves precisely the outstanding supercritical fractional-order question; it does not assert that the scaled errors are monotone in N there, nor that the fixed-pair constant equals $-2g$ there.

2.4 A convenient rational certificate for the universal constant

Proposition 2.4 (An elementary rational error budget). *The sharp universal constant satisfies $C_* < 57/50$. Consequently*

$$E_N - \frac{57}{50} 2^{-N} < g_{a,b} < E_N \quad (a, b > 0, N \geq 1).$$

Proof. The existing axis-maximum theorem gives $b_* \in (1, 2)$. By its positive Mellin representation, $M(b) = \Gamma(b)C(b)$ is log-convex. Also

$$(\log \Gamma)''(b) = \sum_{j \geq 0} \frac{1}{(j+b)^2} \leq \frac{\pi^2}{6} < \frac{5}{3} \quad (1 \leq b \leq 2).$$

On an interval $[u, u+h] \subset [1, 2]$, the error of linear interpolation of $\log \Gamma$ is therefore at most $5h^2/24$. Combining this with the convexity inequality for $\log M$ yields

$$C(b) \leq \max\{C(u), C(u+h)\} \exp(5h^2/24).$$

The accompanying exact standard-library verifier `certify_rational_euler_constant.py` proves

$$C(k/10) < \frac{1137}{1000} \quad (k = 10, 11, \dots, 20).$$

It uses 80 terms of the axis Euler enclosure with error budget $5/4$, which follows directly from the scalar domination above and was also available in the preceding manuscript. Every $n^{-k/10}$ is enclosed on a 2^{-160} grid by integer inequalities: if $r = \lfloor (2^{1600}/n^k)^{1/10} \rfloor$, then $r^{10}n^k \leq 2^{1600} < (r+1)^{10}n^k$. All subsequent operations are exact rational arithmetic. The full rational intervals are saved in `rational_euler_constant_certificate.json`; none of the displayed decimal diagnostics is used for a sign decision. Taking $h = 1/10$ gives

$$C_* < \frac{1137}{1000} e^{1/480} < \frac{1137}{1000} \frac{480}{479} = \frac{13644}{11975} < \frac{57}{50}.$$

The exponential inequality follows from its positive series and $e^x < (1-x)^{-1}$ for $0 < x < 1$. $\square$

Proposition 2.5 (A narrow exact enclosure of the sharp constant). *The unique maximizer and its value satisfy*

$$1.3022165871012412095923717014 < b_* < 1.3022165871012412095923717017,$$

$$1.136561103339509560952586375779942387681723540765044951$$

$$< C_* <$$

$$1.136561103339509560952586375779942387681723540765044952.$$

These are exact rational enclosures, not estimates from a floating-point stationary-point calculation.

Proof. Let L, R be the displayed endpoints for b_* , and put $c = 1.30221658710124120959237170155$. The standard-library verifier `certify_axis_maximum.py` encloses $C(L)$, $C(c)$ and $C(R)$ by exact rational intervals and proves

$$C(c) > C(L), \quad C(c) > C(R).$$

The previously proved strict unimodality implies $L < b_* < R$.

Here are all arithmetic ingredients of that finite certificate. The computation uses 224 axis Euler terms, with the self-contained error budget $5/4$ proved above. To enclose n^{-b} for rational b , write $n = 2^k r$, $1 \leq r < 2$, and use

$$\log n = k \log 2 + 2 \sum_{j=0}^{127} \frac{z^{2j+1}}{2j+1} + \mathcal{R}, \quad z = \frac{r-1}{r+1}, \quad 0 \leq \mathcal{R} \leq \frac{2z^{257}}{257(1-z^2)}.$$

The same formula with $z = 1/3$ encloses $\log 2$. For each resulting $0 \leq x = b \log n < 8$, sum the positive exponential series through $x^{200}/200!$, with tail bounded by

$$\frac{x^{201}}{201!} \frac{1}{1 - x/202},$$

and reciprocate the interval. Every operation is rounded outwards on the dyadic grid 2^{-384} . All sum weights and error budgets are exact rational numbers. The complete rational intervals, along with the strict comparisons above, are recorded in `axis_maximum_certificate.json`.

For the upper enclosure on the maximum, the interpolation argument in Proposition 2.4 gives, with $\delta = 5(R - L)^2/24$,

$$C(c) \leq C_* \leq \max\{C(L), C(R)\} e^\delta < \frac{\max\{C(L), C(R)\}}{1 - \delta}.$$

Applying the rational lower endpoint for $C(c)$ and the two rational upper endpoints at L, R , then rounding the final bounds outwards to 54 decimal places, proves the displayed enclosure. $\square$

3 A uniform saddle expansion for all reflected exponent ratios

Let $f(x) = \log \Gamma(x)$ and $F(x) = -\psi(x)$ on $0 < x < 1$. Both functions are positive. For real $n, m \geq 0$ put

$$M_{n,m} = \int_0^1 f(x)^n f(1-x)^m dx.$$

Reflection gives $M_{n,m} = M_{m,n}$, so it is enough to consider $n \geq m$. The previously proved compact-ratio theorem does not provide a uniform estimate on the entire range $n/m \geq 1$: its saddle in the x coordinate approaches an endpoint. The following construction uses a coordinate and a phase which remain uniformly suitable in that limit.

3.1 The inverse coordinate and the one needed inherited lemma

Write $x(y)$ for the decreasing inverse of f , and define, for $y > 0$,

$$q(y) = f(1 - x(y)), \quad d(y) = \frac{e^y}{F(x(y))}, \quad r(y) = -\frac{yq'(y)}{q(y)}. \quad (9)$$

Thus q is positive and decreasing, while d is positive. Direct differentiation gives

$$r(y) = \frac{1}{J(x(y))}, \quad J(x) = \frac{F(x)f(1-x)}{F(1-x)f(x)}. \quad (10)$$

We use the following established result from the pinned repository.

Lemma 3.1 (Inherited global slope inequality). *The function J is an increasing real-analytic bijection $(0, 1) \rightarrow (0, \infty)$, and $J'(x)/J(x) > 176/225$. Consequently r is an increasing real-analytic bijection $(0, \infty) \rightarrow (0, \infty)$, with $r'(y) > 0$.*

The source is the theorem entitled “Certified global log-gamma slope ratio” in the report *Uniform Transitions and Integral Structures*, at the revision recorded in the provenance appendix. This is an inherited theorem, not a new assertion of this article. Its proof treats the endpoints analytically and proves the compact inequality with 256 exact dyadic interval computations. The dependency is included with its source attribution and freshly replayed in this package. To verify the consequence, differentiate (10), using $x'(y) = -1/F(x(y))$.

We also need elementary quantitative information at infinity. Put

$$c = \frac{\zeta(2)}{2\gamma} - \gamma > 0.$$

The inverse-function theorem applied to $x = e^{-y}\Gamma(1+x)$ and the convergent log-gamma series yield

$$\begin{aligned} x(y) &= e^{-y} - \gamma e^{-2y} + O(e^{-3y}), \\ q(y) &= \gamma e^{-y} \{1 + ce^{-y} + O(e^{-2y})\}, \\ d(y) &= 1 - 2\gamma e^{-y} + O(e^{-2y}), \\ r(y) &= y \{1 + ce^{-y} + O(e^{-2y})\}. \end{aligned} \quad (11)$$

Every fixed number of derivatives may be taken in these expansions. More precisely, the correction functions in $q(y)e^y/\gamma$ and $d(y)$ are analytic functions of e^{-y} near zero. In particular, for each fixed $j \geq 2$,

$$(\log q)^{(j)}(y) = O_j(e^{-y}), \quad d^{(j)}(y) = O_j(e^{-y}), \quad r'(y) \rightarrow 1. \quad (12)$$

The statement for $d^{(j)}$ also holds for $j = 1$. Near zero, $x(y) = 1 - y/\gamma + O(y^2)$, so $q(y) = \log(\gamma/y) + o(1)$ and $d(y) \rightarrow 1/\gamma$. Consequently d has positive finite global upper and lower bounds.

3.2 The all-ratio theorem

Define

$$\Phi_{n,m}(y) = n \log y + m \log q(y) - y. \quad (13)$$

The last term incorporates the exponential part of the Jacobian; keeping it in the phase is essential when m is small. The exact change of variables is

$$M_{n,m} = \int_0^\infty e^{\Phi_{n,m}(y)} d(y) dy. \quad (14)$$

Theorem 3.2 (Uniform expansion without a ratio restriction). *For every real $n > 0$ and $m \geq 0$ there is exactly one positive solution $s = s(n, m)$ of*

$$n = s + mr(s). \quad (15)$$

It is the unique global maximizer of (13), and

$$H = -\Phi''_{n,m}(s) = \frac{1 + mr'(s)}{s} > 0.$$

For each fixed integer $K \geq 0$, as $n \rightarrow \infty$,

$$M_{n,m} = e^{\Phi_{n,m}(s)} d(s) \sqrt{\frac{2\pi}{H}} \left\{ \sum_{j=0}^K \frac{L_j(n, m)}{n^j} + O_K(n^{-K-1}) \right\}, \quad 0 \leq m \leq n. \quad (16)$$

The error constant is independent of m , and each coefficient $L_j(n, m)$ is bounded on $n \geq 2$, $0 \leq m \leq n$. Here $L_0 = 1$. With derivatives evaluated at s ,

$$\frac{L_1}{n} = \frac{d''}{2dH} + \frac{d'\Phi^{(3)}}{2dH^2} + \frac{\Phi^{(4)}}{8H^2} + \frac{5(\Phi^{(3)})^2}{24H^3}. \quad (17)$$

The leading factor can equivalently be written

$$\frac{s^n q(s)^m}{F(x(s))} \sqrt{\frac{2\pi s}{1 + mr'(s)}}. \quad (18)$$

Thus reflection supplies an expansion for every pair $n, m \geq 0$ as $\max(n, m) \rightarrow \infty$.

Proof. First,

$$\Phi'_{n,m}(y) = \frac{n - y - mr(y)}{y}.$$

The function $h_m(y) = y + mr(y)$ increases strictly from zero to infinity. The derivative of the phase is therefore positive before its unique zero and negative afterwards. At that zero its second derivative is $-h'_m(s)/s$, proving the existence, uniqueness and curvature assertions.

The issue is uniformity, rather than this stationary calculation. Choose $s_0 > 0$ so small that $s_0 < 1/2$ and $r(s_0) < 1/2$. For $n \geq 2$, $0 \leq m \leq n$, the saddle equation precludes $s \leq s_0$: otherwise $n = s + mr(s) < 1/2 + n/2 < n$. Thus $s > s_0$ uniformly. On $[s_0/2, \infty)$, positivity, continuity and (11) give constants $c_0, C_0, c_1, C_1 > 0$ such that

$$c_0 \leq r(y)/y \leq C_0, \quad c_1 \leq r'(y) \leq C_1. \quad (19)$$

It follows from the saddle equation that

$$n \asymp (m+1)s, \quad H \asymp n/s^2,$$

with absolute constants throughout the stated parameter range.

For $y, s \geq s_0/2$, the mean-value estimate $h'_m \geq c(m+1)$ gives a global phase bound. Integrating $\Phi' = (h_m(s) - h_m(y))/y$ on either side of s yields

$$\Phi(y) - \Phi(s) \leq -c_2 n \{y/s - 1 - \log(y/s)\} \quad (y \geq s_0/2) \quad (20)$$

for one constant $c_2 > 0$. This calculation has the same sign on both sides: if $y < s$, integrate the positive derivative from y to s . The portion $0 < y < s_0/2$ is also harmless. The phase is increasing there, so its integral is at most a constant times $e^{\Phi(s_0/2)}$. In (20), $(s_0/2)/s \leq 1/2$; hence this contribution is exponentially small relative to $e^{\Phi(s)}s/\sqrt{n}$.

Now put $u = y/s$ and

$$\varphi(u) = \frac{\Phi(su) - \Phi(s)}{n}, \quad b(u) = \frac{d(su)}{d(s)}, \quad A = -\varphi''(1) = \frac{s(1 + mr'(s))}{n}.$$

The number A is bounded above and bounded away from zero. Every fixed derivative of φ and b is bounded uniformly on $1/2 \leq u \leq 3/2$. For example, for $j \geq 2$,

$$\varphi^{(j)}(u) = (-1)^{j-1}(j-1)!u^{-j} + \frac{m}{n}s^j(\log q)^{(j)}(su),$$

and the second term is bounded by (12), compactness on bounded s , and $\sup_{s \geq s_0} s^j e^{-s/2} < \infty$. The same argument bounds the derivatives of b .

These estimates give a parameter-independent neighborhood of $u = 1$ on which $\varphi(u) \leq -c_3(u - 1)^2$. Outside that neighborhood, (20) gives an exponentially small integrable tail. For clarity, integrability on the unbounded part follows from the linear growth of $u - 1 - \log u$ as $u \rightarrow \infty$; near zero it follows from its growth like $-\log u$.

Here is a more explicit remainder argument. Put $u = 1 + v/\sqrt{n}$ and first restrict $|v| \leq n^\epsilon$, with a positive ϵ chosen sufficiently small for the desired fixed order K . On this interval, the cubic and all higher phase terms are uniformly small relative to the quadratic term. Taylor-expand the amplitude and the exponential of those higher terms, retaining all powers through $n^{-K-1/2}$. Using, for example, phase and amplitude derivatives through order $4K+8$ and then decreasing ϵ if necessary bounds the omitted integrand by $C_K n^{-K-1}(1 + |v|^{D_K})e^{-c_4 v^2}$ for a fixed integer D_K . The uniform derivative bounds established above justify these Taylor remainders. The region $|v| > n^\epsilon$ within the common neighborhood is smaller than every power of n^{-1} by the quadratic gap; the preceding global bound handles the rest. All retained odd powers integrate to zero after extension to the whole real line, whose error is again smaller than every power. This proves the stated uniform $O_K(n^{-K-1})$ normalized remainder. Boundedness of the coefficient functions follows from the derivative bounds and the lower bound on A .

For an explicit finite prescription, set $\kappa_j = s^j \Phi^{(j)}(s)/n$ and $b_j = s^j d^{(j)}(s)/d(s)$. If Z is centered Gaussian of variance A^{-1} , then L_j is the coefficient of t^{2j} in

$$\mathbb{E} \left[\left(\sum_{r \geq 0} \frac{b_r Z^r t^r}{r!} \right) \exp \left\{ \sum_{r \geq 3} \frac{\kappa_r Z^r t^{r-2}}{r!} \right\} \right]. \quad (21)$$

Each coefficient uses finitely many terms. The degree-two coefficient gives (17); substitution of $d = e^s/F(x(s))$ gives (18). $\square$

3.3 What uniformity does and does not imply

The theorem includes balanced moments, fixed positive exponent ratios, the logarithmic transition, fixed reflected exponent, and the axis $m = 0$. No lower bound on m is needed. If m is fixed, then

$$s = \frac{n}{m+1} + O_m(n e^{-n/(m+1)}),$$

and the leading factor in (16) reduces to the leading Stirling approximation of $\gamma^m \Gamma(n+1)/(m+1)^{n+1}$. In the ideal limiting kernel $q(y) = \gamma e^{-y}$, $d(y) = 1$, the first coefficient is exactly $L_1 = 1/12$, independently of m . This recovers the first Stirling correction and checks the Jacobian and normalization.

The error in the theorem is uniform relative error of algebraic order in the larger exponent. It does not claim the exponentially small accuracy of the fixed- m residue expansion. Likewise, the inherited sharp least-term remainder concerns a different approximation. These are complementary theorems with distinct error scales.

There is a useful probabilistic consequence. Give Y the density proportional to $e^{\Phi_{n,m}(y)} d(y)$. Uniformly for $0 \leq m \leq n$,

$$\sqrt{H}(Y - s) \implies N(0, 1) \quad (n \rightarrow \infty).$$

Precisely, for each fixed bounded continuous h and $Z \sim N(0, 1)$,

$$\sup_{0 \leq m \leq n} \left| \mathbb{E}h(\sqrt{H}(Y - s)) - \mathbb{E}h(Z) \right| \rightarrow 0.$$

This follows from the same scaled Gaussian domination and uniform tightness. The variance scale is H^{-1} ; unlike the original x -coordinate scale, it remains valid when the exponent ratio tends to infinity. Higher local expansions follow from the coefficient prescription, while a quantitative global normal-distance bound would require an explicitly stated metric and is left for future work.

4 Effective uniform saturation and a growing-index zero regime

The fixed-index concentration theorem in the manuscript gives an existential derivative-order threshold and confines its expansion to a compact interval of the scale a/k . The estimates below remove both restrictions for a leading-order location theorem. They give an explicit, deliberately conservative threshold in terms of the Laurent index alone. The proof treats the smallest and largest zero on the same footing.

Retain the reciprocal-gamma polynomials and the positive Lerch weight

$$\frac{e^{xz}}{\Gamma(1+z)} = \sum_{n \geq 0} Q_n(x) \frac{z^n}{n!}, \quad W_\rho(t) = \frac{1}{1 - \rho e^{-t}}, \quad 0 \leq \rho \leq 1.$$

For $k \geq 1$, write

$$F_{n,k}^\rho(a) = (-1)^{n+k} D_{n,k}^\rho(a) = \int_0^\infty t^k e^{-at} W_\rho(t) Q_n(\log t) dt. \quad (22)$$

Here $D_{n,k}^\rho = \partial_a^k \ell_n(\rho, a)$ for $\rho < 1$ and $D_{n,k}^1 = \gamma_n^{(k)}(a)$, with

$$\ell_n(\rho, a) = \sum_{m \geq 0} \frac{\rho^m \log^n(a+m)}{a+m} = (-1)^n \partial_s^n \Phi(\rho, s, a)|_{s=1}.$$

The integral representation and the global bound of at most n positive zeros, counted with multiplicity, are existing results of the manuscript. We use them explicitly; the new argument concerns quantitative separation and uniform sign stability.

Order the roots of Q_n decreasingly as $x_{n,1} > \cdots > x_{n,n}$ and put $c_{n,j} = e^{-x_{n,j}}$.

Theorem 4.1 (Explicit saturation and simultaneous location bounds). *For every integer $n \geq 2$, define*

$$\delta_n = (3n)^{-(n-2)}, \quad K_n = 576n^2(3n)^{2n-4} = (24n/\delta_n)^2. \quad (23)$$

For every integer $k \geq K_n$ and every $0 \leq \rho \leq 1$, $D_{n,k}^\rho$ has exactly n positive zeros. All are simple. If they are ordered increasingly as $a_{n,1,k}^\rho < \dots < a_{n,n,k}^\rho$, then

$$\left| \log \frac{a_{n,j,k}^\rho}{kc_{n,j}} \right| < \frac{8n}{\sqrt{k}}, \quad 1 \leq j \leq n. \quad (24)$$

Thus the relative bounds

$$e^{-8n/\sqrt{k}}kc_{n,j} < a_{n,j,k}^\rho < e^{8n/\sqrt{k}}kc_{n,j}$$

hold simultaneously for every branch and every deformation parameter. No compactness assumption on the collection of slopes $c_{n,j}$ is made.

The threshold is an upper bound, not a proposed sharp value. In particular, the existing index-two theorem gives saturation already at $k = 1$, whereas (23) gives $K_2 = 2304$. The value of the bound is its uniformity and explicit dependence on n .

4.1 An explicit separation bound for the Appell roots

For a real-rooted polynomial, its *mesh* is the minimum gap between consecutive roots, counted with multiplicity. A multiple root gives mesh zero. During a temporary repeated-root stage, we separately refer to the minimum gap between its *distinct* roots.

We need the classical fact that $p \mapsto p + cp'$ does not decrease mesh. This is the theorem of Riesz recorded as Theorem 1 of Brändén–Krasikov–Shapiro [6]. The following proof also explains why it applies to the infinite reciprocal-gamma product.

Lemma 4.2 (Preservation of mesh). *If p is real-rooted and $c \in \mathbb{R}$, then*

$$\text{mesh}(p + cp') \geq \text{mesh}(p).$$

Proof. The operator $T = 1 + c\partial_x$ preserves real-rootedness: away from the distinct roots r_i of multiplicities m_i , its new roots solve $\sum_i m_i/(x - r_i) = -1/c$. This sum is strictly decreasing on each complementary interval. Its inherited roots have multiplicity $m_i - 1$. The assertion is immediate when $c = 0$ or p has mesh zero.

Suppose the mesh of p is $d > 0$. For any $0 < h < d$, the roots of $p(x)$ and $p(x - h)$ interlace. The elementary real-rooted-pencil criterion says that two real-rooted polynomials of the same degree interlace if and only if every real linear combination is real-rooted or zero. One proof divides one polynomial by the other: the interlacing case has residues of one sign, which makes the rational quotient strictly monotone between poles. Conversely, if every pencil member is real-rooted, that quotient cannot take a real value in the upper half-plane; its imaginary part has constant sign there, forcing the residues, and hence the interlacing order. Common roots follow by cancellation or a limiting argument.

Apply T to this pencil. Since T preserves real-rootedness and commutes with translation, every real combination of $Tp(x)$ and $Tp(x - h)$ is real-rooted or zero. The criterion gives interlacing of this shifted pair, which is equivalent to $\text{mesh}(Tp) \geq h$. Let $h \uparrow d$. $\square$

Lemma 4.3 (Quantitative separation while removing multiplicities). *Let p be real-rooted of degree $n \geq 2$, with at least two distinct roots separated by at least d , where $0 < d \leq 1$. If $c \geq 1/n$, the distinct roots of $p + cp'$ are separated by at least $d/(3n)$.*

Proof. Write the distinct roots as $r_1 < \dots < r_q$, with multiplicities m_i . There is one new root s_i in each gap (r_i, r_{i+1}) , one new root $s_0 < r_1$, and the inherited roots of reduced multiplicity. In an interior gap put $u = s_i - r_i$ and $v = r_{i+1} - s_i$.

If $u < d/2$, every root to the right is at distance at least $d/2$. In the equation $p'(s_i)/p(s_i) = -1/c$, the positive part includes m_i/u , and the magnitude of the negative part is at most $2n/d$. Consequently $u \geq d/(2n)$. This same lower bound is immediate if $u \geq d/2$. If $v < d/2$, the positive part is at most $2n/d$, so the negative part, which includes m_{i+1}/v , is at most $2n/d + 1/c$. Thus in every case

$$u \geq \frac{d}{2n}, \quad v \geq \frac{d}{2n + d/c} \geq \frac{d}{3n}.$$

For the exterior root, $\sum_i m_i/(r_i - s_0) = 1/c$ implies $r_1 - s_0 \geq cm_1 \geq 1/n$. Every possible pair of consecutive distinct new or inherited roots is therefore separated by at least $d/(3n)$. $\square$

Proposition 4.4 (A fully explicit Appell mesh bound). *For every $n \geq 2$,*

$$\text{mesh}(Q_n) \geq \delta_n = (3n)^{-(n-2)}. \quad (25)$$

Proof. Start with x^n and apply $1 + \partial_x$. The result $x^{n-1}(x+n)$ has two distinct roots at distance $n \geq 1$. Apply next $1 + \partial_x/j$, for $j = 2, \dots, n-1$. Their coefficients $1/j$ are all at least $1/n$. Lemma 4.3, used $n-2$ times starting with $d = 1$, shows that the distinct-root gaps of

$$B_n(x) = \prod_{j=1}^{n-1} (1 + \partial_x/j) x^n$$

are at least $(3n)^{-(n-2)}$. At each step inherited multiplicities drop by one and all new roots are simple. After these $n-1$ steps, every root is simple, so this is a bound for the ordinary mesh of B_n .

For $N \geq n-1$ the reciprocal-gamma finite-product polynomial is

$$Q_{n,N}(x) = \left(\prod_{j=1}^N (1 + \partial_x/j) \right) (x + \gamma - H_N)^n.$$

Translations preserve mesh. The remaining factors, with $j \geq n$, do not decrease it by Lemma 4.2. Hence every $Q_{n,N}$ has mesh at least δ_n . The Weierstrass product

$$\Gamma(1+z)^{-1} = e^{\gamma z} \prod_{j \geq 1} (1 + z/j) e^{-z/j}$$

converges locally uniformly. Thus $Q_{n,N} \rightarrow Q_n$ coefficientwise. Continuity of the ordered real roots of monic polynomials of fixed degree preserves the gap inequalities in the limit. This proves (25), including strict simplicity of the limit. $\square$

Lemma 4.5 (The inherited global zero bound). *For $n \geq 0$, $k \geq 1$, and $0 \leq \rho \leq 1$, $F_{n,k}^\rho$ has at most n positive zeros, counted with multiplicity.*

Proof. We recall the short variation argument so that the completeness claim does not depend on a root scan. If a real kernel g has N sign changes, its Laplace transform has at most N positive zeros, provided the exponentially weighted polynomial moments exist. For $N = 0$ the transform has a strict constant sign. For $N > 0$, choose a sign-change point t_0 . Multiplication by $t_0 - t$ removes that change. The new transform is

$$(\partial_a + t_0)G(a) = e^{-t_0 a} \frac{d}{da} (e^{t_0 a} G(a)).$$

By induction it has at most $N - 1$ zeros. Any finite collection of M zeros of G , counted with multiplicity, forces at least $M - 1$ zeros of the displayed derivative by Rolle's theorem with multiplicities. Thus $M \leq N$. Induction also precludes an identically zero transform.

In (22), the factor $t^k W_\rho(t)$ is positive and $Q_n(\log t)$ has n simple sign changes by Proposition 4.4, with the cases $n = 0, 1$ immediate. The bound $W_\rho(t) \leq W_1(t) \leq 1 + t^{-1}$ gives all required integrability near zero, and e^{-at} controls infinity. $\square$

4.2 A scale-free positive-weight estimate

The decisive improvement over a compact-scale Taylor estimate is the following uniform control of the weight on the logarithmic coordinate.

Lemma 4.6 (Logarithmic variation of the Lerch weight). *For $x, y > 0$ and $0 \leq \rho \leq 1$,*

$$\left| \log \frac{W_\rho(xy)}{W_\rho(x)} \right| \leq |\log y|. \quad (26)$$

Proof. Direct differentiation gives

$$\frac{d \log W_\rho(x)}{d \log x} = -\frac{x\rho}{e^x - \rho}.$$

Its magnitude is at most one: since $e^x \geq 1 + x$ and $0 \leq \rho \leq 1$, $e^x - \rho \geq 1 + x - \rho \geq \rho x$. Integrate this derivative between $\log x$ and $\log x + \log y$. The proof is valid at both deformation endpoints. $\square$

Let Y_k have gamma density

$$\frac{k^{k+1}}{k!} y^k e^{-ky}, \quad y > 0.$$

This is shape $k + 1$ and rate k . Keeping the shape shift is important: for example $\mathbb{E}Y_k^{-1} = 1$ exactly.

Lemma 4.7 (An elementary exponential log-moment bound). *For positive integers m, k with $k \geq 16m^2$,*

$$\mathbb{E}(e^{m|\log Y_k|} - 1) \leq \left[2(e^{4m^2/k} - 1) \right]^{1/2} \leq \sqrt{2/3} < 1. \quad (27)$$

Proof. For a positive number y , according to whether $y \geq 1$ or $y < 1$, the square of $e^{m|\log y|} - 1$ is one of $(y^m - 1)^2$ and $(y^{-m} - 1)^2$. Cauchy–Schwarz therefore bounds the square of the left side by

$$\mathbb{E}Y_k^{2m} + \mathbb{E}Y_k^{-2m} - 2\mathbb{E}Y_k^m - 2\mathbb{E}Y_k^{-m} + 2.$$

The integer gamma moments are

$$\mathbb{E}Y_k^r = \prod_{j=1}^r (1 + j/k), \quad \mathbb{E}Y_k^{-r} = \prod_{j=0}^{r-1} (1 - j/k)^{-1} \quad (r \leq k).$$

Both are at least one. Since $k \geq 4m$,

$$\begin{aligned} \log \mathbb{E} Y_k^{2m} &\leq \frac{m(2m+1)}{k} \leq \frac{3m^2}{k}, \\ \log \mathbb{E} Y_k^{-2m} &\leq \frac{2}{k} \sum_{j=0}^{2m-1} j \leq \frac{4m^2}{k}. \end{aligned}$$

Here $-\log(1-u) \leq 2u$ for $0 \leq u \leq 1/2$ supplies the second line. These estimates prove the first inequality in (27). Finally $4m^2/k \leq 1/4$ and $e^{1/4} \leq (1-1/4)^{-1} = 4/3$ give the displayed rational bound. $\square$

Proposition 4.8 (Uniform relative sign control away from Appell roots). *Fix $n, k \geq 1$, a real s at distance at least $d > 0$ from every root of Q_n , and put $m = \lceil 1 + n/d \rceil$. If $k \geq 16m^2$, then, simultaneously for $0 \leq \rho \leq 1$,*

$$\left| \frac{(ke^{-s})^{k+1} F_{n,k}^\rho(ke^{-s})}{k! W_\rho(e^s) Q_n(s)} - 1 \right| \leq \sqrt{2/3} < 1. \quad (28)$$

In particular, $F_{n,k}^\rho(ke^{-s})$ and $Q_n(s)$ have the same sign.

Proof. The substitution $t = e^s y$ in (22) gives the exact normalized expression

$$\frac{(ke^{-s})^{k+1} F_{n,k}^\rho(ke^{-s})}{k! W_\rho(e^s) Q_n(s)} = \mathbb{E} \left[\frac{W_\rho(e^s Y_k)}{W_\rho(e^s)} \frac{Q_n(s + \log Y_k)}{Q_n(s)} \right].$$

For $u \in \mathbb{R}$, the product over roots gives

$$\left| \frac{Q_n(s+u)}{Q_n(s)} - 1 \right| \leq \prod_{j=1}^n \left(1 + \frac{|u|}{|s - x_{n,j}|} \right) - 1 \leq e^{n|u|/d} - 1.$$

Put $R = W_\rho(e^{s+u})/W_\rho(e^s)$. Lemma 4.6 implies $R \leq e^{|u|}$ and $|R - 1| \leq e^{|u|} - 1$. Consequently

$$\left| R \frac{Q_n(s+u)}{Q_n(s)} - 1 \right| \leq e^{(1+n/d)|u|} - 1 \leq e^{m|u|} - 1.$$

Take expectations and apply Lemma 4.7. This also supplies all needed absolute integrability. $\square$

4.3 Completion of the zero theorem and the joint limit

Proof of Theorem 4.1. Put $\eta = 8n/\sqrt{k}$. By (23), $\eta \leq \delta_n/3$ and $k \geq 256$. The intervals $[x_{n,j} - \eta, x_{n,j} + \eta]$ are disjoint. Each endpoint is at distance at least η from every Appell root. For $d = \eta$ the integer in Proposition 4.8 satisfies

$$m = \left\lceil 1 + \frac{\sqrt{k}}{8} \right\rceil \leq 2 + \frac{\sqrt{k}}{8} \leq \frac{\sqrt{k}}{4},$$

so $k \geq 16m^2$. That proposition transfers the opposite Appell signs at the two endpoints to the normalized Lerch family, uniformly in ρ . The decreasing change of coordinate $a = ke^{-s}$ gives one positive zero in each of the corresponding n disjoint intervals.

The global zero bound counts multiplicity and allows at most n . It therefore excludes every additional positive zero and every multiple zero. The j th interval consequently contains precisely $a_{n,j,k}^\rho$ in the stated ordering. Its endpoints yield (24). $\square$

Corollary 4.9 (A growing Laurent index). *Fix $0 < \alpha < 1/2$. For all sufficiently large integers k , simultaneously for*

$$2 \leq n \leq \alpha \frac{\log k}{\log \log k}, \quad 1 \leq j \leq n, \quad 0 \leq \rho \leq 1,$$

the family has exactly n simple positive zeros and

$$\frac{a_{n,j,k}^\rho}{k e^{-x_{n,j}}} = 1 + o(1). \quad (29)$$

More precisely, the logarithmic error is bounded by $8n/\sqrt{k}$ throughout this range.

Proof. Equation (23) gives $\log K_n = 2n \log n + O(n)$. At the largest permitted n , this is at most $(2\alpha + o(1)) \log k < \log k$; the same bound holds for every smaller n , with finitely many bounded indices absorbed by increasing the lower threshold for k . Thus $k \geq K_n$ throughout the range. Theorem 4.1 applies, and $8n/\sqrt{k} \rightarrow 0$ uniformly there. Exponentiation proves (29). $\square$

Remark 4.10 (A replaceable separation input). The argument actually proves the stronger reusable statement: if any $\delta > 0$ is a proved lower bound for $\text{mesh}(Q_n)$, then the same location bound holds once

$$k \geq \max\{256, (24n/\delta)^2\}.$$

The explicitly constructed δ_n makes this unconditional for every index. Sharper analytic estimates or certified Appell root brackets can be inserted without changing the positive-kernel proof. This is a natural route to much smaller effective thresholds.

4.4 Concrete sufficient thresholds from rational root isolation

The proof of Proposition 4.4 gives the more precise inequality $\text{mesh}(Q_n) \geq \text{mesh}(B_n)$. The latter is a rational-polynomial problem. The accompanying `lerch_verify_effective.py` constructs B_n in exact fraction arithmetic, removes its simple zero at the origin, and isolates every remaining root by a Sturm sequence. If $[\ell_i, u_i]$ are the ordered isolating intervals, then $\min_i(\ell_{i+1} - u_i)$ is a rational lower mesh bound. The exact root count certifies completeness of this finite list.

The table records a subset of the replayed cases $2 \leq n \leq 12$. Its last column is exactly $\lceil \max\{256, (24n/\underline{\delta})^2\} \rceil$. The executable checks both the upper-ceiling inequality and its minimality as an integer ceiling by rational comparisons.

n	Certified $\underline{\delta} \leq \text{mesh}(B_n)$	Sufficient k for every $\rho \in [0, 1]$
2	16383/8192	577
3	3333/4096	7830
4	29421/65536	45729
5	4683/16384	176261
6	6495/32768	527797
7	9557/65536	1327197
8	3663/32768	2950044

These are sufficient thresholds obtained from a common quantitative argument; they are not sharp classifications. The known sharp uniform threshold at $n = 2$ remains $k = 1$. The same replay verifies six exact gamma-moment controls at $k = 16m^2$, for $m = 1, 2, 4, 8, 16, 32$, and rejects a deliberately out-of-hypothesis control at $m = 2, k = 4$. These finite checks audit the implementation. The all-index theorem and its growing-index corollary rest on the preceding proofs.

4.5 The terminal branch at arbitrary index

One further general consequence does not require concentration. It extends the known index-two terminal-branch motion law to any index and derivative order whenever the last positive zero is simple.

Proposition 4.11 (Strict leftward motion of a simple terminal zero). *Fix $n, k \geq 1$. On an interval $I \subset [0, 1)$ suppose the largest positive zero $r(\rho)$ of $F_{n,k}^\rho$ exists, is simple, and is followed by no further positive zero. Then $r'(\rho) < 0$ on the interior of I . At $\rho = 0$ the same conclusion holds for a right derivative if the simple terminal branch persists there. In particular, for $n \geq 2$ and $k \geq K_n$ the last branch is strictly decreasing on $[0, 1]$.*

Proof. For $0 \leq \rho < 1$, absolute convergence gives the resolvent identity

$$\partial_\rho F_{n,k}^\rho(a) = \sum_{h \geq 1} \rho^{h-1} F_{n,k}^\rho(a+h). \quad (30)$$

Indeed, write $F_{n,k}^\rho(a) = \sum_{m \geq 0} \rho^m f(a+m)$, where $f = F_{n,k}^0$, and sum over $h+m$; the coefficient of $\rho^{q-1} f(a+q)$ is q . Every summand on the right of (30), evaluated at $a = r(\rho)$, has the same strict sign as $F_{n,k}^\rho$ just to the right of this last simple zero. That is also the sign of $\partial_a F_{n,k}^\rho(r(\rho))$. Implicit differentiation yields $r' = -F_\rho/F_a < 0$.

In the saturated regime all branches are simple for every $\rho \in [0, 1]$. The uniform compact brackets supplied by (24), continuity of the integral in ρ , and uniqueness inside each bracket give continuity at both endpoints. Strict decrease on the interior therefore extends to the closed interval. No finite ρ -derivative at $\rho = 1$ is asserted here. $\square$

5 Integral diagonalization of an odd-weight shuffle minor

The following is a theorem about an integral relation presentation. It says nothing about arithmetic independence of evaluated periods. Its immediate analytic consequence is a uniformly justified choice of Gaussian double-polylogarithm coordinates at every odd weight.

5.1 The relation matrix and its arithmetic structure

Let $m \geq 1$, set $w = 2m + 1$, and use the convention

$$F_{a,b}(z) = \text{Li}_{a,b}(z, 1) = \sum_{n > r \geq 1} \frac{z^n}{n^a r^b}.$$

The standard same-point integral shuffle gives

$$\text{Li}_p(z) \text{Li}_{w-p}(z) = \sum_{a=1}^{2m} \left\{ \binom{a-1}{p-1} + \binom{a-1}{2m-p} \right\} F_{a,w-a}(z), \quad 1 \leq p \leq m. \quad (31)$$

Binomial coefficients outside their usual range are zero. This holds as a holomorphic identity for $|z| < 1$, and also at $z = i$ with the ordinary convergent boundary values. In particular, the possible outer index one is harmless there: summation by parts applied to $H_{n-1}^{(b)}/n$ gives ordinary convergence, and the corresponding iterated integral has no singular endpoint letter.

Select the columns with odd outer index $a = 2k + 1$, so their inner indices are even. With p, k now zero based, the resulting $m \times m$ matrix is

$$(M_m)_{p,k} = \binom{2k}{p} + \binom{2k}{2m-1-p}, \quad 0 \leq p, k < m. \quad (32)$$

Theorem 5.1 (Integral diagonalization of the shuffle minor). *There are explicitly constructible matrices $U_m, V_m \in \text{GL}_m(\mathbb{Z})$ such that*

$$U_m M_m V_m = \text{diag}(1, 3, 5, \dots, 2m-1). \quad (33)$$

Consequently:

1. $\det M_m = (2m-1)!!$.
2. $\text{coker}_{\mathbb{Z}} M_m \cong \bigoplus_{j=0}^{m-1} \mathbb{Z}/(2j+1)\mathbb{Z}$. This direct-sum display is an integral diagonal form, not generally the ordered Smith invariant-factor display.
3. The least positive integer L for which every entry of LM_m^{-1} is an integer is $L_m = \text{lcm}(1, 3, 5, \dots, 2m-1)$.
4. Over characteristic two the matrix has full rank. Over a field of odd characteristic ℓ its rank is

$$m - \left\lfloor \frac{2m + \ell - 1}{2\ell} \right\rfloor.$$

Proof. Put $n = 2m - 1$. The palindromic polynomials

$$Q_k(x) = (1+x)^{2k}(1+x^{n-2k}), \quad 0 \leq k < m,$$

have degree n , and their coefficients of x^p , $0 \leq p < m$, are precisely $(M_m)_{p,k}$. Define

$$b_j(x) = x^j(1+x)^{n-2j}, \quad 0 \leq j < m, \quad B_{p,j} = \binom{n-2j}{p-j}.$$

The coefficient matrix B is lower triangular with diagonal entries one, hence belongs to $\text{GL}_m(\mathbb{Z})$.

Let $E_r(t)$ be the polynomial defined by

$$E_r(y(1-y)) = y^r + (1-y)^r.$$

Its existence with integer coefficients follows, for example, from $E_0 = 2$, $E_1 = 1$, and $E_r = E_{r-1} - tE_{r-2}$. Let E be the $m \times m$ integer matrix whose column j is the coefficient vector of $E_{2j+1}(t)$ in $1, t, \dots, t^{m-1}$. Under $y = x/(1+x)$, $t = x/(1+x)^2$, one has

$$Q_k(x) = (1+x)^n E_{n-2k} \left(\frac{x}{(1+x)^2} \right).$$

If R is the column-reversal matrix, this identity gives the exact factorization

$$M_m = BER. \quad (34)$$

It remains to diagonalize E over the integers. Use the monic odd polynomials

$$p_j(y) = y \prod_{r=1}^j (y^2 - r^2), \quad 0 \leq j < m,$$

and the integer polynomials

$$q_j(t) = (-1)^j \prod_{r=1}^j (t + r(r-1)).$$

Their respective leading coefficients are 1 and $(-1)^j$. Let P be the coefficient matrix of p_j in $y, y^3, \dots, y^{2m-1}$, and let Q be the coefficient matrix of q_j in $1, t, \dots, t^{m-1}$. Both are integer triangular matrices with unit diagonal up to signs, so both are unimodular.

The central-factorial finite-difference identity is

$$\begin{aligned} p_j(y) + p_j(1-y) &= p_j(y) - p_j(y-1) \\ &= (2j+1) \prod_{r=1-j}^j (y-r) \\ &= (2j+1)(-1)^j \prod_{r=1}^j (y(1-y) + r(r-1)) \\ &= (2j+1)q_j(y(1-y)). \end{aligned}$$

For the middle equality, write $p_j(y) = (y+j)^{2j+1}$ and use $z^n - (z-1)^n = n(z-1)^{n-1}$; this is a polynomial identity and needs no restriction on y . Since substitution $t \mapsto y(1-y)$ is injective on $\mathbb{Z}[t]$, comparison in the t basis gives

$$EP = QD, \quad D = \text{diag}(1, 3, \dots, 2m-1).$$

Combining this with (34),

$$U_m = (BQ)^{-1}, \quad V_m = RP$$

are integral unimodular matrices and satisfy (33).

The determinant sign is also fixed by these formulas: $\det B = \det P = 1$, while $\det Q = \det R = (-1)^{m(m-1)/2}$. Thus $\det M_m = \det D > 0$. The cokernel and field-rank assertions follow from integral equivalence. For odd ℓ , the number of positive odd multiples of ℓ not exceeding $2m-1$ is the displayed floor.

Finally, $M_m^{-1} = V_m D^{-1} U_m$, with integral U_m, V_m and integral inverses. Therefore LM_m^{-1} is integral if and only if LD^{-1} is integral, if and only if every $2j+1$ divides L . This proves the optimal denominator assertion. $\square$

Smith factors without ambiguity. For each prime ℓ , sort the m integers $v_\ell(1), v_\ell(3), \dots, v_\ell(2m-1)$ into $e_{\ell,1} \leq \dots \leq e_{\ell,m}$. The ordered Smith invariant factors are exactly $d_r = \prod_\ell \ell^{e_{\ell,r}}$. For $m=5$, for example, they are $(1, 1, 1, 3, 315)$, while the determinant is 945 and the optimal inverse denominator is 315. At $m=8$ they are $(1, 1, 1, 1, 1, 3, 15, 45045)$. Thus determinant size and the required common denominator of the inverse are quite different quantities.

5.2 The uniform odd-inner-index basis

Corollary 5.2 (An integral justification for the discovery basket). *In the rational vector space on the $2m$ formal double coordinates $F_{a,2m+1-a}$, modulo the m same-point shuffle product rows, the m classes*

$$F_{2m,1}, F_{2m-2,3}, \dots, F_{2,2m-1}$$

form a basis. Before taking the quotient, every double is an explicit linear combination of these odd-inner-index coordinates and products $\text{Li}_p \text{Li}_{2m+1-p}$, with all coefficients in $L_m^{-1}\mathbb{Z}$.

Proof. Separate (31) into even-inner coordinates X and odd-inner coordinates Y . If N is the integer coefficient matrix of the latter, the equation is $M_m X + NY = P$, where P is the product vector. Hence $X = M_m^{-1}P - M_m^{-1}NY$. The theorem proves both existence and the coefficient-ring assertion. It also proves that the m rows have rank m , so the m surviving coordinates are independent in this specified formal quotient. Additional functional equations or numerical evaluation may introduce further relations. $\square$

Relation to Zagier's matrix. Zagier's 2012 paper [7], Section 5, equation (36), defines the $(m-1) \times (m-1)$ matrix

$$(A_m)_{r,s} = \binom{2m-2s}{2r-1} + \binom{2m-2s}{2m-2r}, \quad 1 \leq r, s < m.$$

It is permutation-equivalent to the minor of M_m obtained by removing row zero and column zero: reverse the columns, and fold the odd row indices $2r-1$ using $p \leftrightarrow 2m-1-p$. These folded row indices cover $1, \dots, m-1$ exactly once. The first column of M_m is $(1, 0, \dots, 0)^T$, so integer column subtraction removes its first row's other entries. Therefore

$$A_m \text{ is integrally equivalent to } \text{diag}(3, 5, \dots, 2m-1).$$

In particular $|\det A_m| = (2m-1)!!$, the determinant formula conjectured in Zagier's equation (37); its sign is determined by the explicit permutations. The paper's inverse formula was proved by Ding Ma [8] and subsequently by Yawen Ma and Lee-Peng Teo [9], *Another Proof of Zagier's Matrix Conjecture*, Journal of Integer Sequences 25 (2022), Article 22.6.4. We do not claim priority for the determinant identity or for the known inverse formula. The direct integral diagonalization above supplies all Smith factors and the exact denominator statement in the manuscript's shuffle coordinates.

5.3 A short exact weight-eleven identity

The arithmetic theorem supplies concrete identities as well as a basis count. A useful example has a four-row certificate.

Corollary 5.3 (A weight-eleven Gaussian double reduction).

$$\begin{aligned} & 105g_{9,2} + 210g_{10,1} + 35g_{8,3} - 7g_{6,5} + 5g_{4,7} - 7g_{2,9} \\ &= \frac{223339}{29727129600}\pi^{11} + \frac{1785}{131072}\beta(2)\zeta(9) + \frac{63}{512}\beta(4)\zeta(7) \\ &\quad - \frac{15}{64}\beta(6)\zeta(5) - \frac{21}{16}\beta(8)\zeta(3). \end{aligned}$$

Proof. At general z the stronger functional identity is

$$\begin{aligned} & 105F_{9,2}(z) + 210F_{10,1}(z) + 35F_{8,3}(z) - 7F_{6,5}(z) + 5F_{4,7}(z) - 7F_{2,9}(z) \\ &= -7\text{Li}_2(z)\text{Li}_9(z) + 14\text{Li}_3(z)\text{Li}_8(z) - 16\text{Li}_4(z)\text{Li}_7(z) + 8\text{Li}_5(z)\text{Li}_6(z). \end{aligned}$$

It is $-7, 14, -16, 8$ times the shuffle rows with $p = 2, 3, 4, 5$, respectively. Direct coefficient comparison gives, in increasing outer-index order, the vector $(0, -7, 0, 5, 0, -7, 0, 35, 105, 210)$. This is the complete small certificate of the functional identity. Specialize to $z = i$, take imaginary parts, and use

$$\operatorname{Re} \operatorname{Li}_k(i) = -2^{-k}(1 - 2^{1-k})\zeta(k), \quad k > 1, \quad \operatorname{Im} \operatorname{Li}_k(i) = \beta(k).$$

The even-zeta and odd-beta evaluations give the displayed rational coefficient of π^{11} and the remaining four products. $\square$

6 Frozen Gaussian candidates at weights eleven and thirteen

Retain $g_{a,b} = \operatorname{Im} F_{a,b}(i)$ and $S_p = \sum_{n \geq 0} (-1)^n H_n / (2n+1)^p$. The following identity is conjectural; the exact matrix theorem above justifies the choice of comparison coordinates, but does not imply membership of the mixed sum in their span.

Conjecture 6.1 (Weight eleven).

$$\begin{aligned} S_{10} = & \frac{570649054}{415912609} g_{10,1} + \frac{30750200}{415912609} g_{8,3} - \frac{14642272}{415912609} g_{6,5} \\ & + \frac{15009920}{415912609} g_{4,7} - \frac{641796160}{1247737827} g_{2,9} + \frac{15870231535729}{2649404577860812800} \pi^{11} \\ & - \frac{852385525}{851789023232} \beta(2)\zeta(9) - \frac{88141815}{7605259136} \beta(4)\zeta(7) \\ & - \frac{437907815}{6654601744} \beta(6)\zeta(5) - \frac{468343007}{1663650436} \beta(8)\zeta(3) - 2\beta(10) \log 2. \end{aligned}$$

In the ordered basket

$$(S_{10}, g_{10,1}, g_{8,3}, g_{6,5}, g_{4,7}, g_{2,9}, \pi^{11}, \beta(2)\zeta(9), \beta(4)\zeta(7), \beta(6)\zeta(5), \beta(8)\zeta(3), \beta(10) \log 2),$$

the frozen primitive integer vector is

$$\begin{aligned} & (2649404577860812800, -3635091082366156800, -195881824419840000, 93272725153382400, \\ & -95614679384064000, 1362768401793024000, -15870231535729, 2651259936960000, \\ & 30705505754112000, 174344763875328000, 745847853554073600, 5298809155721625600). \end{aligned}$$

The search used 1600 finite Euler weights computed as exact integers, 550-digit arithmetic for the values, then PSLQ at 300 working digits, tolerance 10^{-280} , coefficient bound 10^{30} , and at most 30000 iterations. After freezing the vector, the residual evaluated using the retained 550-digit values was approximately 2.575×10^{-464} . This number is a diagnostic and is consistent with the finite Euler truncation. The accompanying exact rational certificate is the authoritative finite proximity claim. Neither small residuals nor a residual interval containing zero prove this conjecture.

An independent 120-digit quadrature audit used

$$S_{10} = -\frac{1}{9!} \int_0^\infty \frac{t^9 e^{-t} \log(1 + e^{-2t})}{1 + e^{-2t}} dt,$$

and

$$g_{a,b} = \frac{1}{(a-1)!} \int_0^1 (-\log x)^{a-1} \operatorname{Im} \frac{i \operatorname{Li}_b(ix)}{1 - ix} dx.$$

For $b = 1$, it instead used the integrated logarithmic formula

$$g_{a,1} = -\frac{1}{2(a-2)!} \int_0^1 \frac{(-\log x)^{a-2} \log(1+x^2) \arctan x}{x} dx.$$

The normalized residual was approximately 3.49×10^{-121} . This confirms the normalization by a different representation, with floating-point quadrature rather than the finite Euler sums. No rigorous quadrature error estimate is asserted, and this audit is not a premise of the rational proximity proof.

Proposition 6.2 (Certified weight-eleven proximity). *Let R_{10} be S_{10} minus the displayed conjectural right side. Then*

$$-10^{-650} < R_{10} < 10^{-650}.$$

The delivered rational residual interval contains zero. This statement is a proved enclosure for the normalized residual; the conjectural assertion $R_{10} = 0$ remains open.

Proof. Apply the rational bounds proved immediately below with $N = 2200$ Euler terms, 600 terms in each of the two arctangent series, and 800 terms in the logarithm series. The standard-library program `verify_gaussian_proximity.py` reconstructs all twelve basket intervals from these finite integer/rational formulas, reads the frozen primitive vector from `gaussian_candidates.json`, and performs ordinary exact rational interval addition and multiplication. It divides the integer dot product by its first coefficient 2649404577860812800, so the output is precisely R_{10} , rather than the unnormalized integer residual. Direct rational comparison proves that the lower endpoint exceeds -10^{-650} and the upper endpoint is below 10^{-650} . Exact endpoints of this residual and all basket values are in `gaussian_proximity_certificate.json`. No floating-point approximation enters this verification. $\square$

6.1 A further candidate at weight thirteen

The same structured basket yields one additional frozen conjecture. Let

$$\mathbf{b}_{12} = (S_{12}, g_{12,1}, g_{10,3}, g_{8,5}, g_{6,7}, g_{4,9}, g_{2,11}, \pi^{13}, \\ \beta(2)\zeta(11), \beta(4)\zeta(9), \beta(6)\zeta(7), \beta(8)\zeta(5), \beta(10)\zeta(3), \beta(12)\log 2).$$

The conjecture is $\mathbf{c}_{12} \cdot \mathbf{b}_{12} = 0$, where

$$\mathbf{c}_{12} = (67206362357091388096512000, -92814968433857420014387200, \\ -5035645608385269517516800, 2454053342295358816911360, \\ -2557982490705884558131200, 4600624150682630105333760, \\ 10639593135931482596966400, -43411484181262303199, \\ 5190040482548669193600, 71048339137356985958400, \\ 743250995585205475737600, 5397522019420435985203200, \\ 22801241978065356167577600, 134412724714182776193024000).$$

It has greatest common divisor one. A 440-digit PSLQ calculation with tolerance 10^{-420} , coefficient bound 10^{30} , and at most 50000 iterations produced this vector from the 550-digit/1600-Euler-term

values. Its retained-precision integer residual was approximately 2.581×10^{-457} . The same exact rational calculation as for S_{10} proves

$$\left| \frac{\mathbf{c}_{12} \cdot \mathbf{b}_{12}}{67206362357091388096512000} \right| < 10^{-650}.$$

The delivered normalized interval contains zero. This remains a conjectural identity, supported by a proved proximity enclosure.

The first, weaker search at 300 digits and tolerance 10^{-280} returned a different vector. Its higher-precision residual failed immediately. That discarded vector is preserved in `gaussian_candidates.json`, separately from the retained vector. Using the same rigorous basket intervals, its *unnormalized* integer residual R_{bad} satisfies

$$-\frac{2191}{10^{277}} < R_{\text{bad}} < -\frac{2190}{10^{277}} < 0.$$

Thus the original weight-thirteen vector is rigorously false, not merely unconfirmed. The exact endpoints are included in the certificate and the optional `-s12` replay verifies both the retained proximity statement and this rejection. This is a practical example of why discovery precision must not be reused as validation precision.

6.2 The exact rational error bound used by the certificate

For clarity, the error bound is rederived here. It is the direct one-sided Euler bound already used in the pinned manuscript, not a new proof of the proposed identity. For a sequence f_n , write

$$\Delta^k f_0 = \sum_{n=0}^k (-1)^n \binom{k}{n} f_n, \quad E_N(f) = \sum_{k=0}^{N-1} \frac{\Delta^k f_0}{2^{k+1}}.$$

The exact finite identity

$$E_N(f) = \frac{1}{2^N} \sum_{n=0}^{N-1} (-1)^n \left(\sum_{r=n+1}^N \binom{N}{r} \right) f_n \quad (35)$$

follows by interchanging the two finite sums and using Pascal's identity. It is convenient computationally because its weights are integers divided by one power of two; the implementation does not form an unstable floating-point finite-difference triangle.

For $a \geq 2$ and integer $b \geq 1$, put $f_n = H_{2n}^{(b)} / (2n+1)^a$. The elementary Mellin formulas give

$$f_n = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^1 \int_0^1 \frac{(-\log x)^{a-1} (-\log t)^{b-1}}{1-t} ((x^2)^n - (x^2 t^2)^n) dt dx.$$

Each grouped difference is integrable, including at $b = 1$; the individual terms should not be separated there. Finite differences turn the bracket into $(1-x^2)^k - (1-x^2 t^2)^k$, which is nonpositive when $k \geq 1$. Summing the geometric tail yields the exact remainder expression

$$g_{a,b} - E_N(f) = \frac{2^{-N}}{\Gamma(a)\Gamma(b)} \int_0^1 \int_0^1 \frac{(-\log x)^{a-1} (-\log t)^{b-1}}{1-t} \left\{ \frac{(1-x^2)^N}{1+x^2} - \frac{(1-x^2 t^2)^N}{1+x^2 t^2} \right\} dt dx.$$

One justification is to start with the convergent original series: for $a \geq 2$ it is absolutely summable since $b \geq 1$. The displayed integral equals its sum by dominated convergence of grouped kernels.

Alternatively, the uniform bound below makes the series of Euler remainders absolutely integrable and identifies the same sum.

The function $B_N(u) = (1-u)^N/(1+u)$ decreases on $[0, 1]$ and obeys $|B'_N(u)| \leq N+1$. Thus the bracket lies between $-(N+1)x^2(1-t^2)$ and zero. Integrating this bound gives

$$E_N(f) - \frac{N+1}{2^N 3^a} (1+2^{-b}) \leq g_{a,b} \leq E_N(f). \quad (36)$$

The integral of $x^2(1-t^2)$ against the positive Mellin weight is exactly $f_1 = H_2^{(b)}/3^a$, explaining every factor in the bound. For $f_n = H_n/(2n+1)^p$, $p \geq 2$, replace $x^2 t^2$ by $x^2 t$. The same argument gives

$$E_N(f) - \frac{N+1}{2^N 3^p} \leq S_p \leq E_N(f). \quad (37)$$

The single-value sequences $(2n+1)^{-s}$ and $(n+1)^{-s}$ are Hausdorff moment sequences with mass one. Their omitted Euler tails are nonnegative and bounded by 2^{-N} . This encloses $\beta(s)$ and $\eta(s)$; the relation $\zeta(s) = \eta(s)/(1-2^{1-s})$ encloses every odd zeta value needed. The elementary constants are evaluated independently with

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239), \quad \log 2 = 2 \sum_{n \geq 0} \frac{1}{(2n+1)3^{2n+1}}.$$

The alternating arctangent series is enclosed by its first omitted term. For M terms in the logarithm series, the positive tail is at most $9/[4(2M+1)3^{2M+1}]$. Machin's identity uses the real principal arctangent: the tangent of $4 \arctan(1/5) - \arctan(1/239)$ is one, and this angle lies in $(0, \pi/2)$, fixing the branch.

All these bounds are rational. The certificate applies (35) using an integer common denominator $D = \text{lcm}(1, \dots, 2N)$, and performs only integer and rational interval operations. It encloses every basket value before forming the fixed linear combination. Thus it does not assume that the conjectural residual is zero at any stage.

7 Critical review, corrections, and limits of the conclusions

The pinned manuscript already contains important corrections to earlier stages of this project. Its S_4 reduction is a theorem, its constant-one Euler enclosure has a stated domain, and several low-index Lerch motion heuristics have already been replaced by proved phase information. Those corrections should survive integration. The present review found no reason to retract those proved results. The changes proposed below update open items that the new arguments settle and prevent stronger interpretations that the arguments do not support.

7.1 The Gaussian maximum now controls every Euler index

The caution in the caption of the Gaussian-axis figure in `05-subcritical-stieltjes.tex`, and the corresponding paragraph in `10-discovery.tex`, distinguish a first-remainder maximum from a universal Euler constant. That distinction was justified: knowing the largest value of $-2g_{a,b}$ alone does not bound the later scaled tails. Lemma 2.1 supplies the missing comparison. The replacement should state Theorem 2.2 and Corollary 2.3, including the strict positive-order inequality and the limiting equality at $a = 0$.

The existing counterexample to a universal constant one remains valid. The improved theorem changes the constant to the optimal $C_* > 1$; it does not extend the old constant-one theorem with its hypotheses removed. Similarly, the stronger bounds already proved on smaller parameter regions should remain available.

Proposition 7.1 (Scaled Euler errors need not decrease). *For the positive integer pair $(a, b) = (2, 1)$,*

$$R_2(2, 1) - R_1(2, 1) > \frac{1}{60} > 0.$$

Nevertheless $R_N(2, 1) \rightarrow 0$. Thus, even at integer orders, the largest scaled error need not occur at $N = 1$.

Proof. The first relevant coefficients are $c_3 = 1/6$, $c_5 = 1/12$, and $c_7 = 1/20$. Hence $d_1 = -1/6$, $d_2 = -1/4$, $d_3 = -3/10$, and

$$E_1 = 0, \quad E_2 = -\frac{1}{24}, \quad E_4 = -\frac{11}{120}.$$

The strict one-sided enclosure gives $g_{2,1} < E_4$. Therefore

$$R_2 - R_1 = 4E_2 - 2E_1 - 2g_{2,1} = -\frac{1}{6} - 2g_{2,1} > \frac{1}{60}.$$

The limit follows from Theorem 2.2. This is a finite symbolic counterexample, independent of a numerical plot or a floating-point evaluation of $g_{2,1}$. $\square$

For a fixed positive pair, the positive sequence R_N tends to zero, so its supremum is attained at at least one finite index. Determining that index and its dependence on (a, b) is a separate problem from the sharp constant uniform in both orders.

7.2 Uniformity must be attached to an approximation and a domain

The former compact-ratio moment theorem should retain its original statement. Theorem 3.2 supplies an additional expansion whose relative remainder is uniform on $0 \leq m \leq n$. The two phases place different parts of the Jacobian in the exponential, and their saddles need not agree exactly at finite exponents. Asymptotic equivalence does not justify identifying those saddles term by term.

The all-ratio theorem gives any prescribed algebraic order in n^{-1} . It does not replace exponentially accurate fixed- m residue expansions, or their least-term remainder estimates, by a bound of the same strength. A future matched theorem must state the approximation and error scale explicitly in each overlap.

For Lerch zeros, the new threshold K_n proves completeness by sign transfer and a multiplicity bound. A numerical scan would not suffice. The threshold should always be described as sufficient. At index two the source already proves saturation for every $k \geq 1$, much stronger than either of our general-purpose quantitative bounds. The new result is useful because it applies to arbitrary n , arbitrary branches, and the full closed deformation interval simultaneously.

7.3 Formal coordinates and numerical periods

The quotient in Corollary 5.2 is defined by the specified m shuffle rows. Its dimension is exactly m by an invertible minor. Evaluating at i , or imposing additional functional equations, may create

further relations. Thus neither this dimension nor the Smith form proves linear independence of the corresponding real constants.

The determinant of the related double-zeta matrix and its inverse have published antecedents [7, 8, 9]. The central-factorial construction is presented as a direct proof and an arithmetic extension in these coordinates; exhaustive worldwide priority for its consequences has not been established. A repository integration should preserve this attribution instead of marking the known determinant as a newly settled open conjecture.

7.4 The mixed-color obstruction remains

The weight-seven S_6 identity remains unresolved. The existing coordinate change between $(g_{6,1}, g_{4,3}, g_{2,5})$ and $(g_{6,1}, g_{5,2}, g_{4,3})$ uses two shuffle rows and is already proved in the source. Its restricted-presentation obstructions likewise have their own exact meanings. Integral diagonalization controls the single-color doubles; it does not supply the missing mixed-color functional equation.

A full weight-seven imaginary-word presentation would have 24514 admissible conjugation coordinates in the scoped convention. No completed rank computation or reduction certificate for that enlarged presentation is asserted here. This size is a planning datum, not an independence result or an explanation that the conjecture must be true.

The retained S_{10} and S_{12} relations are also conjectures. Their exact proximity intervals contain zero, so they cannot decide equality. The rejected first S_{12} vector, by contrast, has an interval strictly separated from zero and is rigorously false. The archive keeps the rejected search record for reproducibility and identifies the retained vectors in a separate canonical input file.

8 Computational evidence and reproducibility

The calculations serve three distinct purposes: exact finite certification, independent numerical diagnostics, and exploratory identity discovery. All proof-relevant interval decisions use integers and rational endpoints. The moment diagnostics and Mellin audit use arbitrary-precision floating-point arithmetic and are labeled accordingly. The article’s symbolic proofs establish the all-parameter statements.

8.1 What was verified

Computation	Coverage and outcome	Logical role
Euler kernel checks	2601 rational scalar cases, 2025 strict interior cases, and 560 exact kernel/tail identities pass.	Implementation audit of an analytic theorem.
Universal constant	Eleven rational-order grid enclosures imply $C_* < 57/50$; a separate 384-bit outward interval calculation certifies the narrow b_* and C_* brackets.	Exact finite certificates, using the inherited unique-maximum theorem.
Global gamma slope	All 256 dyadic subintervals replay successfully; the smallest lower bound exceeds $7/4$.	Reproduction of the inherited compact part of the slope proof.

Computation	Coverage and outcome	Logical role
Moment expansion	24 cases at 60 working decimal digits cover balanced, fixed-ratio, square-root, fixed, fractional, axis, and transition families.	Independent quadrature diagnostics.
Lerch mesh	Exact Sturm isolation for $2 \leq n \leq 12$; six gamma-moment positive controls and one rejected negative control.	Finite improved thresholds and implementation audit.
Shuffle matrices	Exact factorizations, unimodularity, determinants, Smith factors and denominator checks for $1 \leq m \leq 20$; the short weight-eleven row identity also passes.	Finite audit of the all- m integer proof.
Gaussian candidates	Both retained normalized residuals lie strictly inside $(-10^{-650}, 10^{-650})$; the rejected S_{12} vector is separated from zero.	Exact proximity and exact rejection, respectively.
Independent S_{10} audit	120-digit Mellin quadrature gives a normalized residual about 3.49×10^{-121} .	Independent normalization check; no rigorous quadrature error asserted.

8.2 Interpreting the diagnostic figures

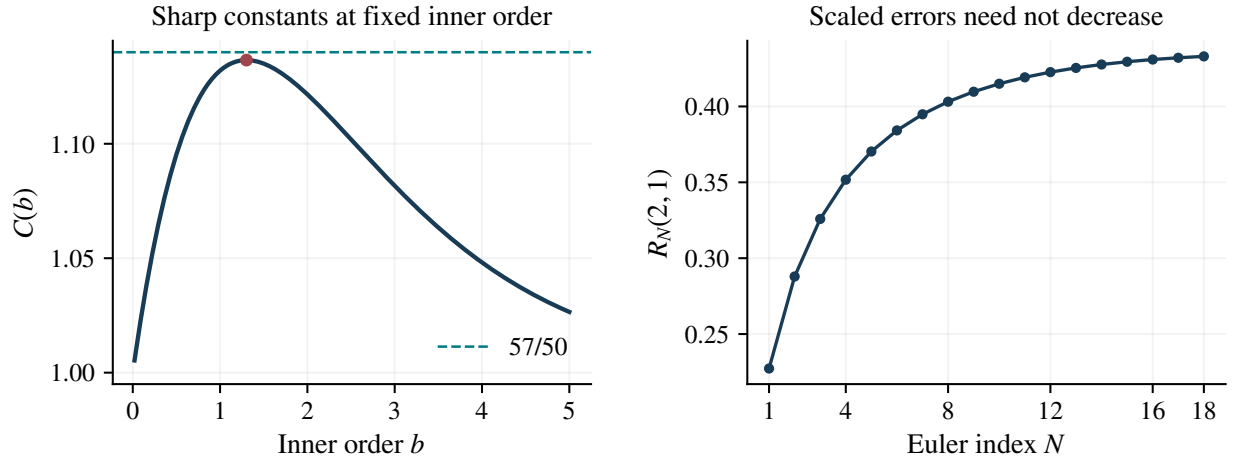


Figure 1: Left: the inner-order constant $C(b)$ and its unique maximum, with the convenient rational budget $57/50$. Right: a finite diagnostic plot of the scaled errors $R_N(2,1)$. Its initial increase is certified symbolically by Proposition 7.1; the plot is not used to prove a global shape or to identify its eventual maximum. The curves are numerical illustrations of proved bounds.

For example, for $m = n = 20, 80, 320$, the corrected relative errors have absolute values approximately 4.39×10^{-5} , 1.95×10^{-6} , and 1.08×10^{-7} . For $m = n/4$ the corresponding values are approximately 7.41×10^{-4} , 4.98×10^{-5} , and 2.89×10^{-6} . These finite values illustrate the theorem but cannot determine the uniform error constant. The axis diagnostic also checks against the source's strict factorial comparison; it does not incorrectly substitute $M_{n,0} = \Gamma(n+1)$ as an exact identity.

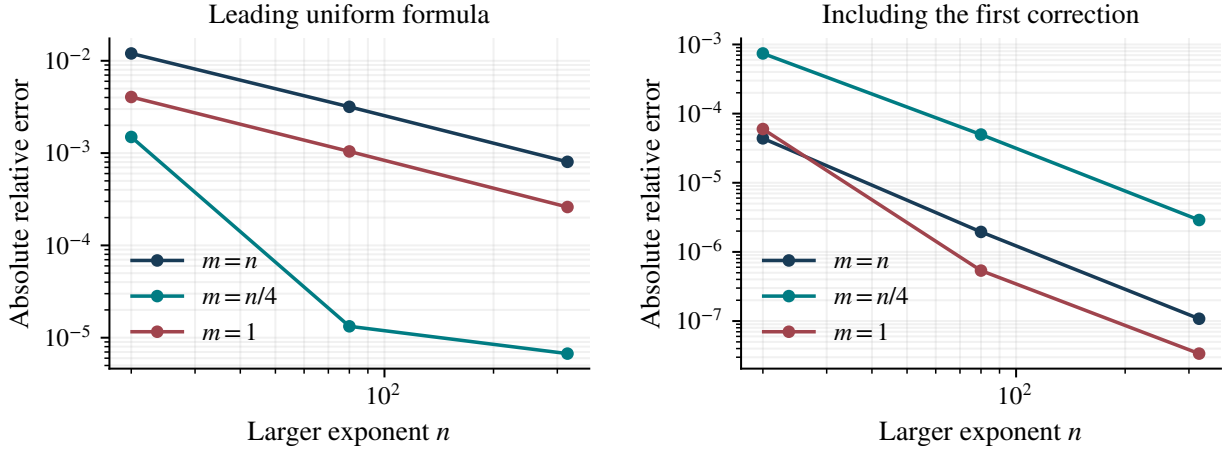


Figure 2: Absolute relative errors in the all-ratio moment approximation, before and after the first correction, for three of the seven tested parameter families. The reference values are computed by quadrature in $t = -\log x$, independent of the inverse-coordinate saddle construction. The first correction generally improves accuracy at sufficiently large exponents; cancellation can make an uncorrected finite case unusually accurate, so pointwise improvement at every input is not a theorem.

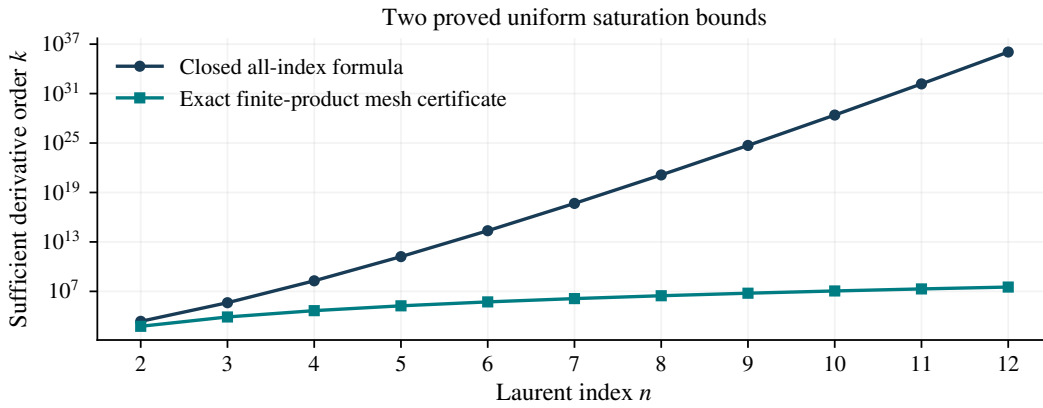


Figure 3: Two sufficient Lerch saturation thresholds: the explicit all-index formula and the improved bounds obtained from exact finite Sturm isolation of B_n . The vertical axis is logarithmic. The displayed integers are consequences of the same proved sign-transfer criterion; they are not estimates of the sharp transition. In particular the known index-two threshold is 1, below both curves.

8.3 A portable replay protocol

The archive provides `code/verify_all.py`. The default command runs the exact certificates and finite matrix audit. It records commands, exit status, elapsed time, and output hashes in a replay manifest. A `--numerical` option additionally recomputes the 24 moment diagnostics; the independent 120-digit Mellin audit has its own optional command. No verification command requires network access or a checkout of the source repository. The inherited gamma-slope program is copied unchanged and explicitly attributed.

The Gaussian input is `code/gaussian/gaussian_candidates.json`. The historical discovery files retain both the unsuccessful first weight-thirteen search and the successful higher-precision search. Their role is audit history; the verifier reads only the canonical retained/rejected records. This avoids inadvertently promoting an obsolete integer vector when integrating the work.

LaTeX compilation uses `code/build_article.py` to concatenate the preamble, ordered section fragments and bibliography into a complete `article.tex`, then `latexmk-pdfarticle.tex`. The final article source can be compiled without Python when the bundled figure PDFs are present. Figure regeneration is optional and uses the saved diagnostic data. The archive includes a file-hash manifest and the exact source revision for every inherited dependency.

9 Further research questions and a proposed programme

The strongest next steps are those that convert the new structural information into a sharper theorem or a finite exact identity certificate. The following questions are separated by their mathematical obstruction and the kind of evidence that would count as progress.

9.1 Sharp fixed-pair Euler errors

For fixed $a, b > 0$, determine

$$C(a, b) = \max_{N \geq 1} R_N(a, b)$$

and the set of maximizing indices. Existence follows from positivity and $R_N \rightarrow 0$, but the maximizer can occur after the first remainder. Does the sequence have at most one local maximum outside the parameter regions where monotonicity is already proved? Are there curves in the (a, b) plane across which the maximizing integer changes? A useful first goal is an analytic classification of the sign of $R_{N+1} - R_N$ using the two-variable positive probability kernel.

The limit $a \downarrow 0$ at fixed b gives $C(a, b) \rightarrow C(b)$ by the first-index lower bound and the uniform upper bound. Determine a uniform first correction in a , and whether the maximizing index is eventually 1 on compact b intervals. Joint limits where a or b approach zero while N grows may reveal additional transition scales. They cannot be inferred from the fixed-pair limit $R_N \rightarrow 0$ alone.

The scalar proof suggests a broader approximation problem: characterize families f whose fixed-midpoint divided differences are monotone and whose endpoint secants yield optimal multiplicative-increment bounds. For Euler-type transformations, this could produce universal constants for positive kernels beyond the harmonic double polylogarithms.

9.2 Effective all-ratio moment asymptotics

The theorem proves the existence of a remainder constant for each fixed order. Produce explicit numerical constants valid for all $0 \leq m \leq n$ above a stated n_0 . This requires quantitative bounds on

the normalized phase and amplitude derivatives, plus the global phase-gap constant. The existing interval proof of the slope inequality provides a concrete starting point rather than an abstract compactness argument.

Can the inherited monotonicity of J be proved by a short analytic inequality, eliminating the compact interval computation? Ordinary convexity of $\log \Gamma$ is insufficient, as the source already explains. A successful proof would make the all-ratio theorem fully analytic and could sharpen its uniform curvature bounds.

Find a uniform approximation retaining the exponentially small fixed- m corrections while also handling the balanced region. The current theorem controls every algebraic order, but matching it to the residue expansion requires analysis of the correction $q(y)e^y/\gamma - 1$ when me^{-s} changes scale. One target is an explicit overlap theorem that recovers both expansions with their own natural error budgets.

The probabilistic corollary invites quantitative questions. Establish a uniform Kolmogorov or total-variation error bound for $\sqrt{H}(Y - s)$, then a uniform Edgeworth expansion with tail control. The metric, centering and normalization must be fixed before a rate is claimed. Related questions concern logarithmic derivatives of $M_{n,m}$, which encode the mean and covariance of $\log f(x)$ and $\log f(1 - x)$ under the tilted moment measure.

9.3 Appell mesh, saturation, and growing zero configurations

The bound $\text{mesh}(Q_n) \geq (3n)^{-(n-2)}$ is deliberately coarse. Determine the actual asymptotic size of the smallest gap. Even a polynomial lower bound $\text{mesh}(Q_n) \gg n^{-A}$ would, through the proved transfer argument, give a polynomial sufficient threshold $k \gg n^{2A+2}$. Such an improvement would extend the growing-index regime far beyond $n = O(\log k / \log \log k)$.

The current proof estimates the product over roots by its worst distance to any root. Replace that bound with local reciprocal-gap sums such as

$$\sum_{\ell \neq j} |x_{n,j} - x_{n,\ell}|^{-1} \quad \text{and} \quad \sum_{\ell \neq j} |x_{n,j} - x_{n,\ell}|^{-2}.$$

Together with centered gamma moments, this may improve the location error $O(n/\sqrt{k})$ and recover constant offsets uniformly for growing n . It would also distinguish extreme roots from interior roots when their local geometries differ.

For each small index, determine the least k that guarantees saturation for every $\rho \in [0, 1]$, and the least k for each global motion law. Certified continuation, exact endpoint analysis and the global multiplicity bound should be used together. The new terminal-branch theorem settles its direction whenever that branch is simple, but it does not order the motion of all interior branches at small k .

More generally, replace W_ρ by a positive family with a uniformly bounded logarithmic derivative. The sign-transfer proof then changes only through that bound. Determine which non-Lerch deformations inherit effective saturation, and which require additional endpoint integrability. This would isolate the root-stability mechanism from its original special-function setting.

9.4 Integral presentations and exact Gaussian identities

The immediate identity target remains a finite rational certificate for the existing S_6 formula. Use the integral shuffle diagonalization to remove redundant coordinates before building Cayley, distribution, conjugation and regularized double-shuffle rows. Each regularized row must carry an

analytic justification. A successful computation should return an exact row combination producing the target, not only a rank or a floating-point null vector.

Conjecture 9.1 (A uniform Gaussian span question). *For every integer $m \geq 1$, the number $S_{2m} + 2\beta(2m) \log 2$ belongs to the rational span of*

$$\{g_{2m-2j, 2j+1} : 0 \leq j < m\} \cup \{\pi^{2m+1}\} \cup \{\beta(2j)\zeta(2m+1-2j) : 1 \leq j < m\}.$$

The proved S_2, S_4 cases in the repository and the conjectural S_6, S_8, S_{10}, S_{12} relations motivate this formulation. Their different evidential status is essential. The all-weight matrix theorem justifies the chosen coordinates but does not prove that a mixed sum belongs to their span. An appropriate generating-function identity could settle many weights at once; until it is found, exact certificates at successive weights are meaningful intermediate goals.

Determine analogous integral diagonal forms at even weights, other roots of unity, and greater depth. The odd-weight proof uses the reflection $y \leftrightarrow 1 - y$ and central factorials in a precise way; it does not automatically furnish a higher-depth presentation. Integral data would identify torsion primes and optimal denominator budgets before expensive exact linear algebra is attempted.

9.5 A practical order of attack

The nearest finite targets are the S_6 row certificate, sharper certified Appell mesh bounds, and explicit constants in the first-order all-ratio moment error. The more structural targets are an analytic global slope proof, a general mixed-Gaussian generating identity, and an asymptotic theory for the Appell mesh. These tasks can proceed in parallel because their proof dependencies are largely separate.

For every proposed identity, freeze the basis and integer vector before validation; recompute it by an independent representation; produce exact residual bounds; and pursue an algebraic or analytic certificate of equality. For every asymptotic claim, state all joint parameter ranges and preserve the distinction between finite diagnostics and a uniform remainder proof. The present package supplies examples of both successful proofs and a rigorously rejected numerical candidate.

A Source provenance and integration map

All comparisons to the repository refer to commit

3447bc59b78d138a7f0d536b66ac6e5cc5d5e49f

in <https://github.com/VladimirReshetnikov/ProveIt>. The commit is dated 10 October 2026 and publishes the reviewed 375-page real-order manuscript milestone. The present article is a separate continuation prepared against that version. The archive's `provenance.json` records hashes of the relevant original files, the inherited verifier, and the provenance of each main claim.

New section	Suggested manuscript location	Status update
Sharp universal Euler bound	After the real-order Euler and Gaussian axis analysis in Chapter 5.	Close the universal-constant question; retain stronger restricted bounds.
All-ratio moments	After Chapter 7, together with the companion report's joint-moment and certified proportional saddle material.	Add an unrestricted-ratio algebraic expansion; retain fixed-order exponential results.
Effective Lerch zeros	After the global variation bound and fixed-index concentration material in Chapter 9.	Replace a purely existential threshold by an explicit sufficient one; add the growing-index regime.
Integral shuffle minor	With the Gaussian reduction and relation-matrix material in Chapters 4–5.	Add the integral presentation and denominator theorem with the Zagier/Ma attribution.
S_{10} and S_{12}	Alongside the frozen S_6 and S_8 candidates.	Mark both as conjectural; attach exact proximity certificates.
Research agenda	Revise the corresponding open items in Chapter 10.	Keep fixed-pair constants, sharp thresholds and mixed reductions open.

The inherited global-slope theorem is sourced from the report `uniform-transition-continuation`, in its section file `sections/04b-proportional-moments.tex`. An unchanged copy is included as `integration/inherited_global_slope.tex`, and its original exact verifier is included under `code/moments/`. The original file includes its endpoint argument, compact certificate, and proportional theorem. It is reference material, not a second new proof or a required input to compiling this article.

For an explicit view of that dependency, write $P = \psi_1$ and

$$\frac{J'}{J} = \frac{F(x)}{f(x)} + \frac{F(1-x)}{f(1-x)} - \frac{P(x)}{F(x)} - \frac{P(1-x)}{F(1-x)}.$$

The source bounds this expression below by $176/225$ on $(0, 1/16]$ using convergent gamma series and elementary inequalities; reflection handles $[15/16, 1)$. Its 256 rational intervals cover $[1/16, 1/2]$ completely. The replayed minimum lower endpoint is

$$\frac{2237344070433247164830307425599}{1267650600228229401496703205376} > \frac{7}{4}.$$

Every rounding is outward on a 2^{-100} grid, and all special constants are enclosed by rational sums with explicit tails. Thus this inherited input is a computer-assisted theorem with a portable exact replay, not a sampled monotonicity assertion.

The supplied integration notes include exact replacement wording for the superseded Euler caution and the research-status entries. They do not propose removing the original hypotheses or

recasting a numerical period relation as a proved identity. No change to the remote repository is part of this deliverable; the modular sections and complete source are ready for review and integration.

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